Confusion Matrix

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

Recall and choose your starting point

> Record your starting explanation for Confusion Matrix before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Scores and Calibration Intuition

Without opening the old lesson, explain one key idea from Scores and Calibration Intuition. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Nearest Neighbors

Without opening the old lesson, explain one key idea from Nearest Neighbors. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Dataset Cards

Without opening the old lesson, explain one key idea from Dataset Cards. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 21

Practise with fading help

> Complete the supported Confusion Matrix trace and record the first step you can explain without help.

Use W2: Explain the two error types for Confusion matrices. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 21

Individual reasoning check

> Check the proposed Confusion Matrix explanation against the supplied facts before discussing it.

Use the worked case for Confusion matrices. Proposed explanation: “Positive means good”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 21

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Changed independent case

On20 cases: **true positive**s5, **false negative**s4, **false positive**s2, **true negative**s9. Build the **confusion matrix** with actual class as rows and predicted class as columns. Compute accuracy.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 21

Your learning target

> Connect the Confusion Matrix target to a trace and a justified conclusion.

Positive=spotted. TP=6,FN=2,FP=3,TN=9. Draw a matrix with actual labels in rows and **prediction**s in columns. Compute total and accuracy.

WORKBOOK · 6 OF 21

Solve the given case

> Solve the supplied Confusion Matrix case and show the deciding evidence.

Positive is spotted. TP=8, FP=2, FN=4, TN=6.

Build actual-row/predicted-column matrix and total.

Show your trace, calculation or **evidence**

Write or draw your reasoning and evidence.

WORKBOOK · 7 OF 21

Check a changed case

Apply the procedure to the changed case and record why the conclusion follows.

Treat fern as positive. Counts are TP=6, FP=2, FN=3 and TN=9. How many examples and how many correct **prediction**s are there?

My independent answer and reason

Write or draw your reasoning and evidence.

Week 15 Test and reflect

WORKBOOK · 8 OF 21

Design a revealing test

> Specify a changed input, expected result and observation separately.

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

Project checkpoint

> Justify what Confusion Matrix adds to Responsible Plant Classifier and keep the evidence.

Publish the Plant Classifier **confusion matrix** with positive-class definition, counts, denominator, and test-set version.

The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

WORKBOOK · 10 OF 21

Self check

> Judge explanation, evidence, transfer and limits separately; attach the work behind your judgement.

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

WORKBOOK · 11 OF 21

Transfer practice from the original programme

> Use this earlier case as transfer practice. A case already discussed with feedback is not fresh assessment evidence.

Positive=spotted. TP=6,FN=2,FP=3,TN=9. Draw a matrix with actual labels in rows and **prediction**s in columns. Compute total and accuracy.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

WORKBOOK · 12 OF 21

W1 Build the count matrix

> Build the count matrix. Apply the supplied Confusion Matrix facts and keep your reasoning.

> **Where to find the supplied example**
>
> Reference for this exercise: open this week's student lesson, "Confusion Matrix", at "See how it works" and "Follow a worked example". Use the original worked values or code named in the question; the later pictorial case is a separate example. Keep this reference beside your workbook while practising.

Use all eight worked IDs. Draw a matrix with reference rows and **prediction** columns, both ordered clear then blocked. Place the IDs in cells before writing the counts.

My reasoning and evidence

WORKBOOK · 13 OF 21

W2 Explain the two error types

> Explain the two error types. Apply the supplied Confusion Matrix facts and keep your reasoning.

Declare blocked positive. Give TP,TN,FP,FN counts and explain V3 and V4 in plain language. Does positive mean a desirable outcome?

My reasoning and evidence

WORKBOOK · 14 OF 21

W3 Audit totals and equal accuracy

Audit totals and equal accuracy. Apply the supplied Confusion Matrix facts and keep your reasoning.

Where to find the supplied example

Reference for this exercise: open this week's student lesson, "Confusion Matrix", at "See how it works" and "Follow a worked example". Use the original worked values or code named in the question; the later pictorial case is a separate example. Keep this reference beside your workbook while practising.

Give the worked row totals, column totals, total count and accuracy. Compare its errors with [[3,0], [3,2]] using the same axes and eight reference records.

My reasoning and evidence

WORKBOOK · 15 OF 21

W4 Count a fresh evaluation independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Count a fresh evaluation independently. Apply the supplied Confusion Matrix facts and keep your reasoning.

Each pair below is reference then **prediction**: R1 clear/clear, R2 clear/blocked, R3 clear/blocked, R4 blocked/blocked, R5 blocked/clear, R6 blocked/blocked. Use clear then blocked on both axes and reference rows.

Build the matrix, give row and column totals, name TP,TN,FP,FN with blocked positive, and calculate accuracy.

My reasoning and evidence

WORKBOOK · 16 OF 21

W5 Interpret a zero count carefully

> Interpret a zero count carefully. Apply the supplied Confusion Matrix facts and keep your reasoning.

A report gives zero **false positive**s but omits the evaluated sample size, axes and class definitions. What information is missing, and why does zero not prove that future false alarms are impossible?

My reasoning and evidence

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Pictorial case practice: Confusion Matrix

> Read the confusion matrix as four distinct stories about a plant decision. Explain the labelled step before checking the changed condition.

Read the **confusion matrix** as four distinct stories about a plant decision.

Positive means spotted. Toy results: TP=4, FN=1, FP=3, TN=12. Rows are actual spotted/healthy; columns predicted spotted/healthy.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-4/15

Positive means spotted

	predicted spotted	predicted healthy
actual spotted	4	1
actual healthy	3	12

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-4/15

Confusion Matrix

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Fill actual spotted row

Actual spotted: [4, 1]. Four caught; one missed.

2. Fill actual healthy row

Actual healthy: [3, 12]. Three false alarms; twelve correctly clear.

3. Compute accuracy

Correct=4+12=16 of total 20, giving80%.

CHANGE ONE THING

A review corrects one missed spotted leaf, moving it from FN to TP.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why is adding TP alone without reducing FN an incorrect revision? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Responsible Plant Classifier record.

Project: Responsible Plant Classifier

Current milestone: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	

Evidence to keep	My record
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

WORKBOOK · 20 OF 21

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Read the **confusion matrix** as four distinct stories about a plant decision.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Correct=4+12=16 of total 20, giving**80%**.

1. Compute accuracy

2. Fill actual spotted row

3. Fill actual healthy row

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Compute accuracy Correct=4+12=16 of total 20, giving**80%**. Read the picture in order. Correct=4+12=16 of total 20, giving**80%**.

The arrows show which recorded **input** or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: A review corrects one missed spotted leaf, moving it from FN to TP. Which conclusion is supported by the supplied facts?

1. Counts become TP 5, FN 0, FP 3, TN 12; accuracy is 17/20=85%.

2. This one example guarantees correct results for every future **input**.

3. We can decide the result without examining the changed **input** or the procedure.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Counts become TP 5, FN 0, FP 3, TN 12; accuracy is 17/20=85%. One decision becomes correct while the total number of evaluated leaves remains20.

Repair a missing building block

Review Scores and Calibration Intuition (ai-4/14). Goal: Interpret calibration using groups of forecasts. Short practice: Does a calibrated 0.8 score mean every such leaf is spotted?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Nearest Neighbors (ai-4/12). Goal: Compute absolute distances and rank **training** neighbours. Short practice: Why is the k=1 outcome at length5 not uniquely determined without a tie rule?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Dataset Cards (ai-4/9). Goal: Inspect row meaning missing values and repeated records. Short practice: What two counts should the revised card show?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

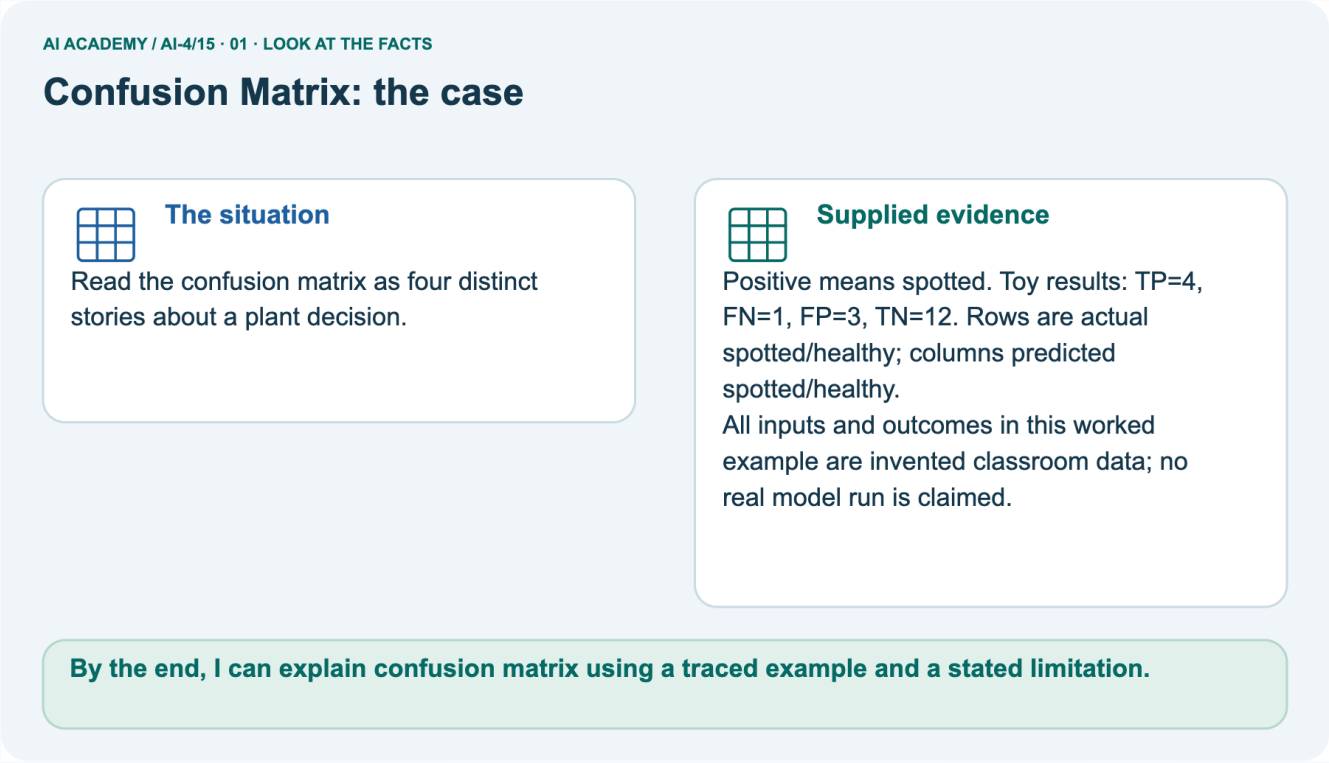
Drafts stay in this browser when storage is available; clear drafts on a shared device.

> **What I repaired, the evidence I kept and the next test**

Visual concept journey

> Explain each Confusion Matrix picture from its labels, then record a relevant limit.

01. Confusion Matrix: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

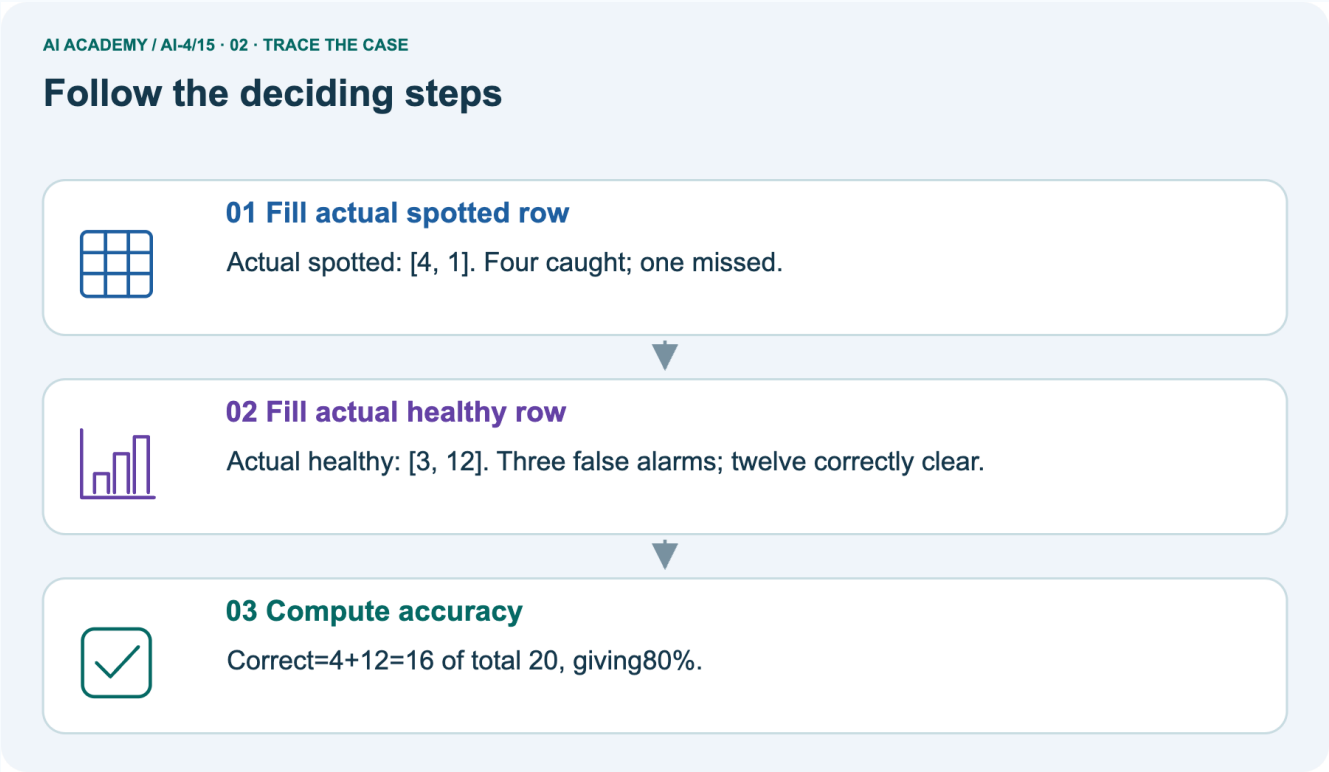
Read the **confusion matrix** as four distinct stories about a plant decision. Supplied facts: Positive means spotted. Toy results: TP=4, FN=1, FP=3, TN=12. Rows are actual spotted/healthy; columns predicted spotted/healthy.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

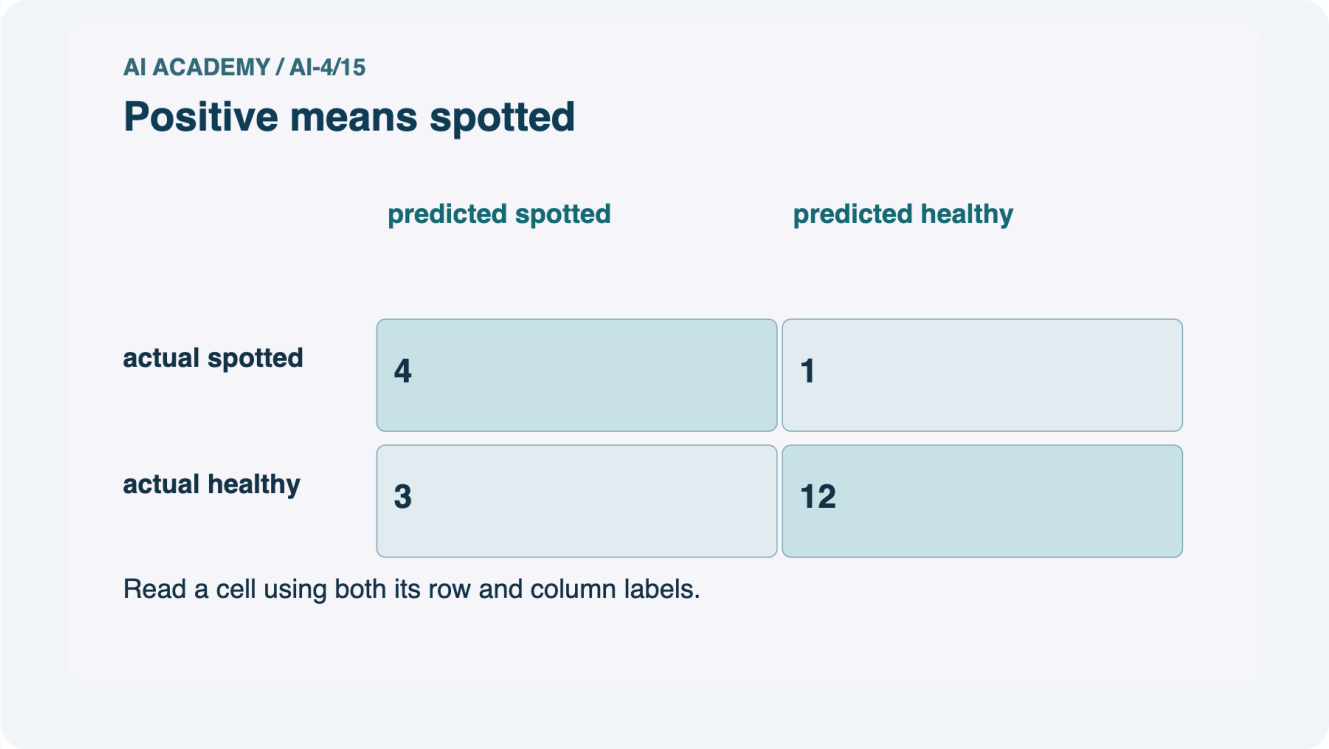
Fill actual spotted row: Actual spotted: [4, 1]. Four caught; one missed. → Fill actual healthy row: Actual healthy: [3, 12].

Three false alarms; twelve correctly clear. → Compute accuracy: Correct=4+12=16 of total 20, giving**80%**.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Positive means spotted



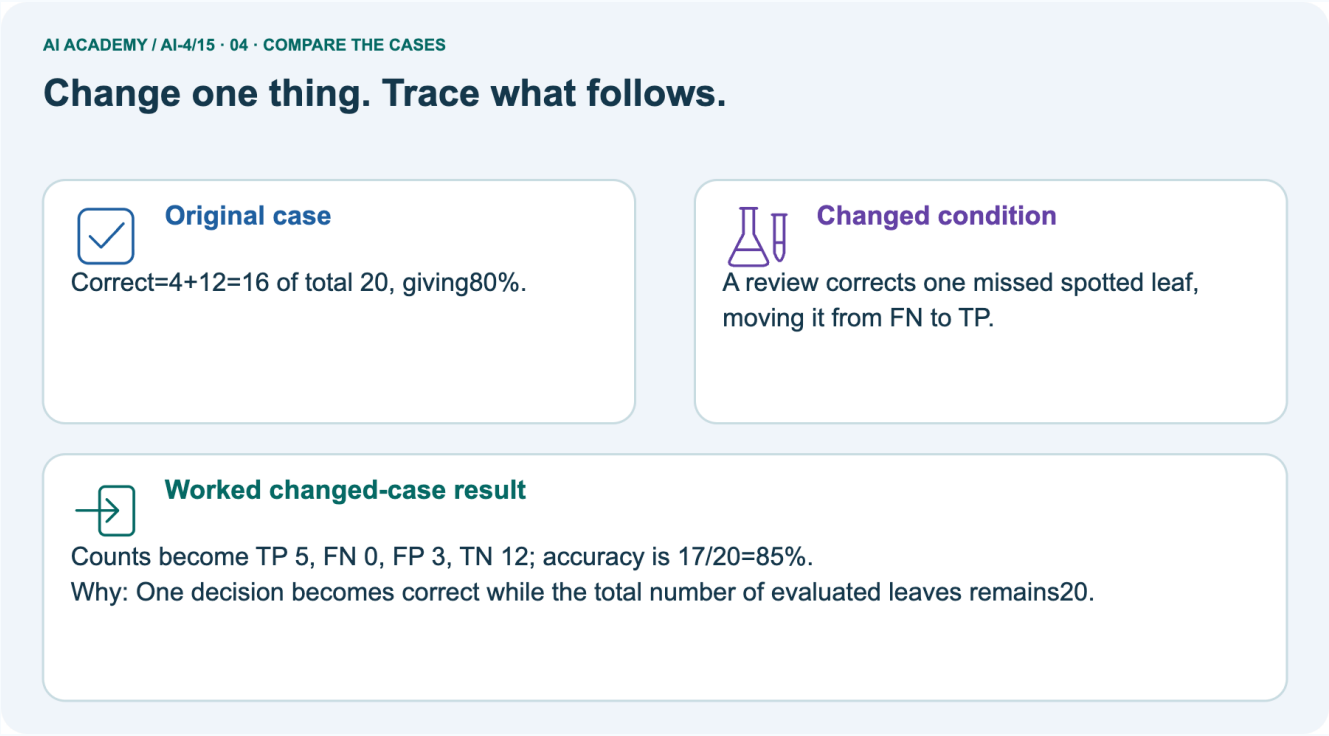
Follow the labelled picture with the matching explanation. Open the vector diagram

Positive means spotted. actual spotted: predicted spotted = 4, predicted healthy = 1. actual healthy: predicted spotted = 3, predicted healthy = 12.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

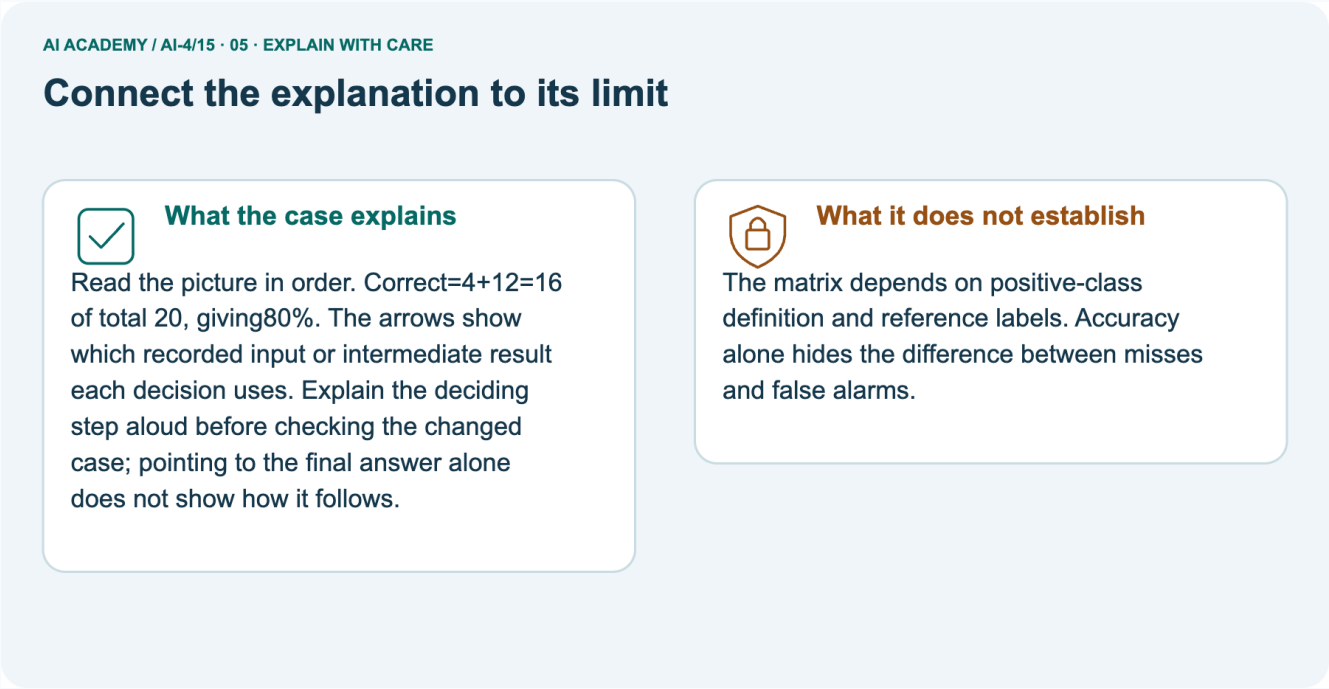
Original case: Correct=4+12=16 of total 20, giving**80%**. Changed condition: A review corrects one missed spotted leaf, moving it from FN to TP. Expected result: Counts become TP 5, FN 0, FP 3, TN 12; accuracy is 17/20=85%.

Reason: One decision becomes correct while the total number of evaluated leaves remains20.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. Correct=4+12=16 of total 20, giving**80%**. The arrows show which recorded **input** or intermediate result each decision uses.

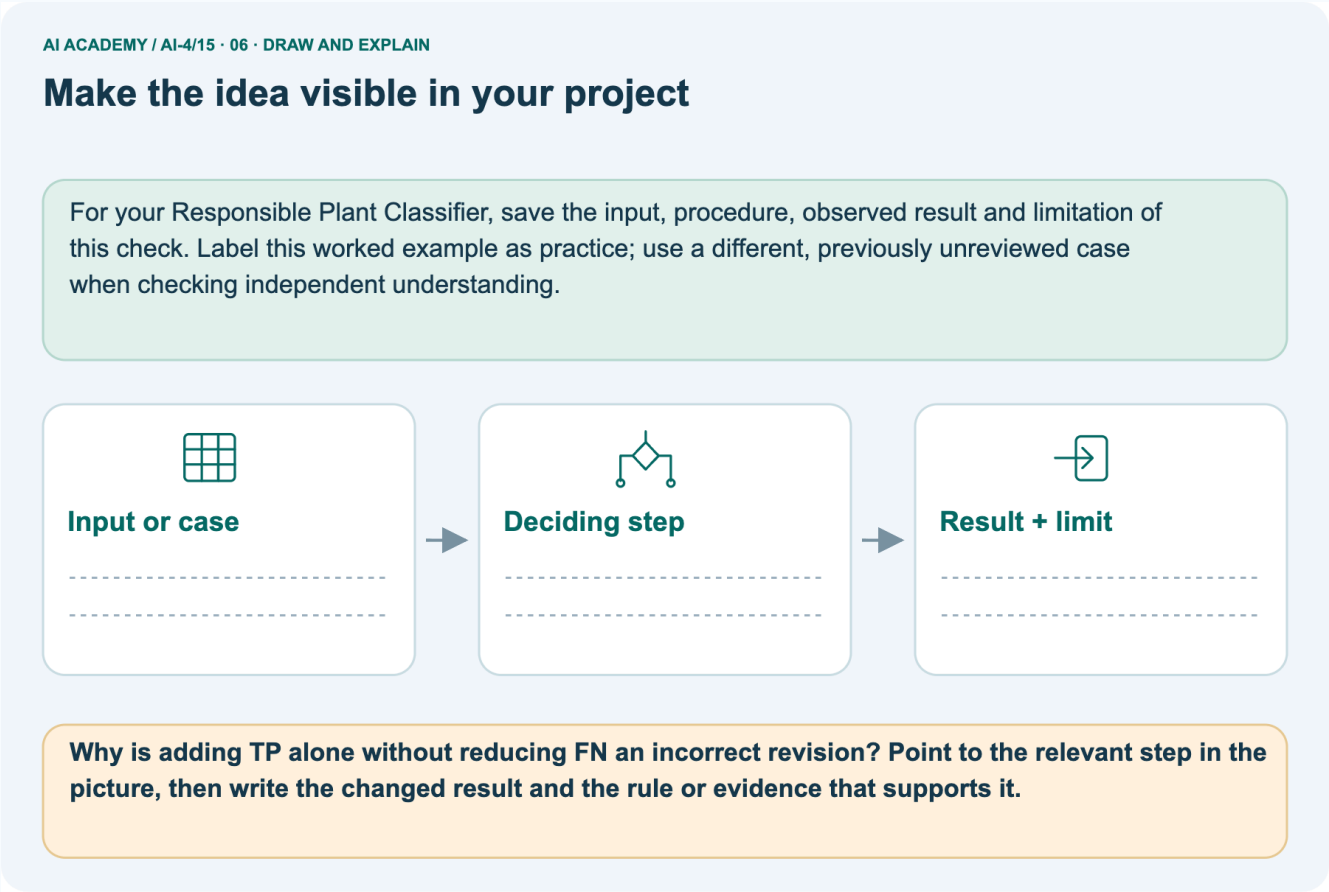
Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows. Limit: The matrix depends on positive-class definition and reference labels.

Accuracy alone hides the difference between misses and false alarms.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Why is adding TP alone without reducing FN an incorrect revision? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.